MANAGEMENT INFORMATION SYSTEMS
ICT2162
LECTURE 03

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Using Information Systems to Achieve Competitive Advantage

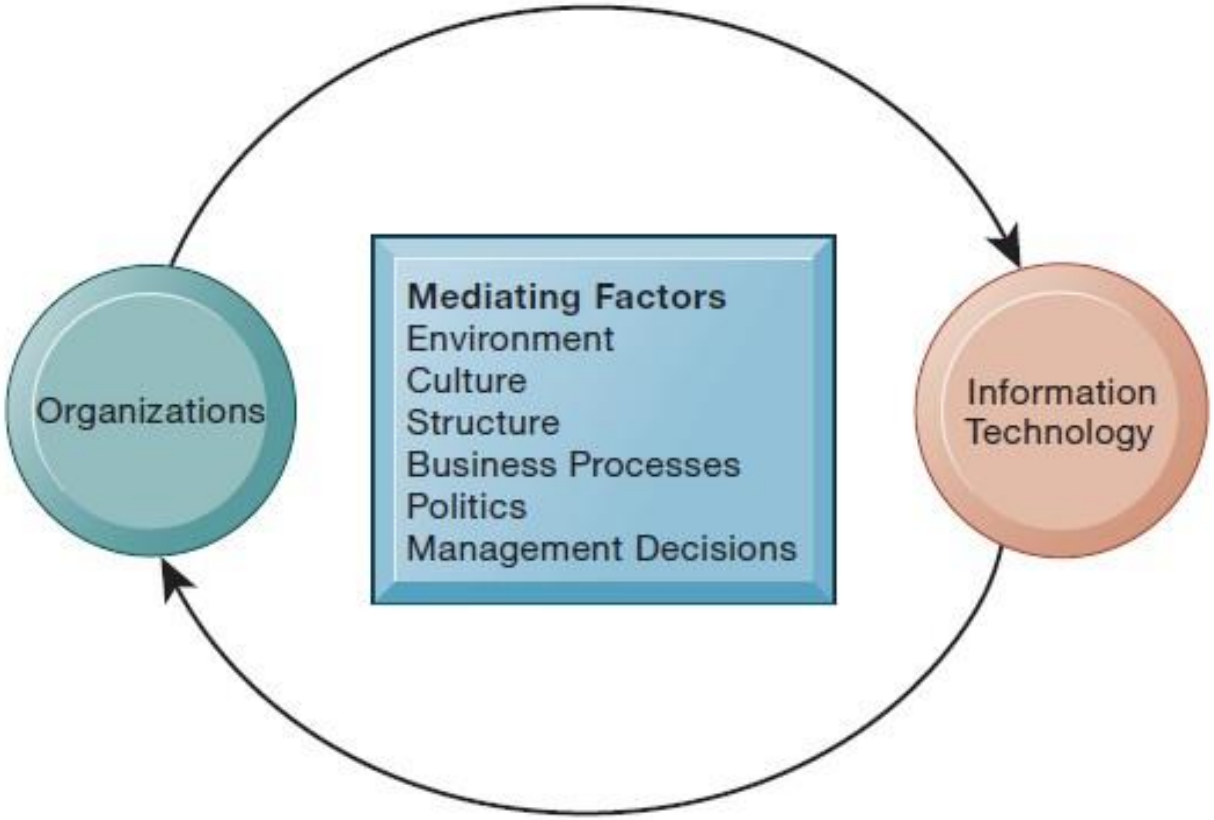
What we discuss today?

1. What is an Organization
2. Features of Organizations
3. Porter's Five Forces Model
4. Porter's Value Chain Analysis

Objectives

1. Which features of organizations do managers need to know about to build and use information systems successfully?
2. What is the impact of information systems on organizations?
3. How do Porter's competitive forces model, the value chain model help companies develop competitive strategies using information systems?
4. What are the challenges posed by strategic information systems, and how should they be addressed?

Relationship Between Organizations and Information Technology



The Two-way Relationship Between Organizations and Information Technology

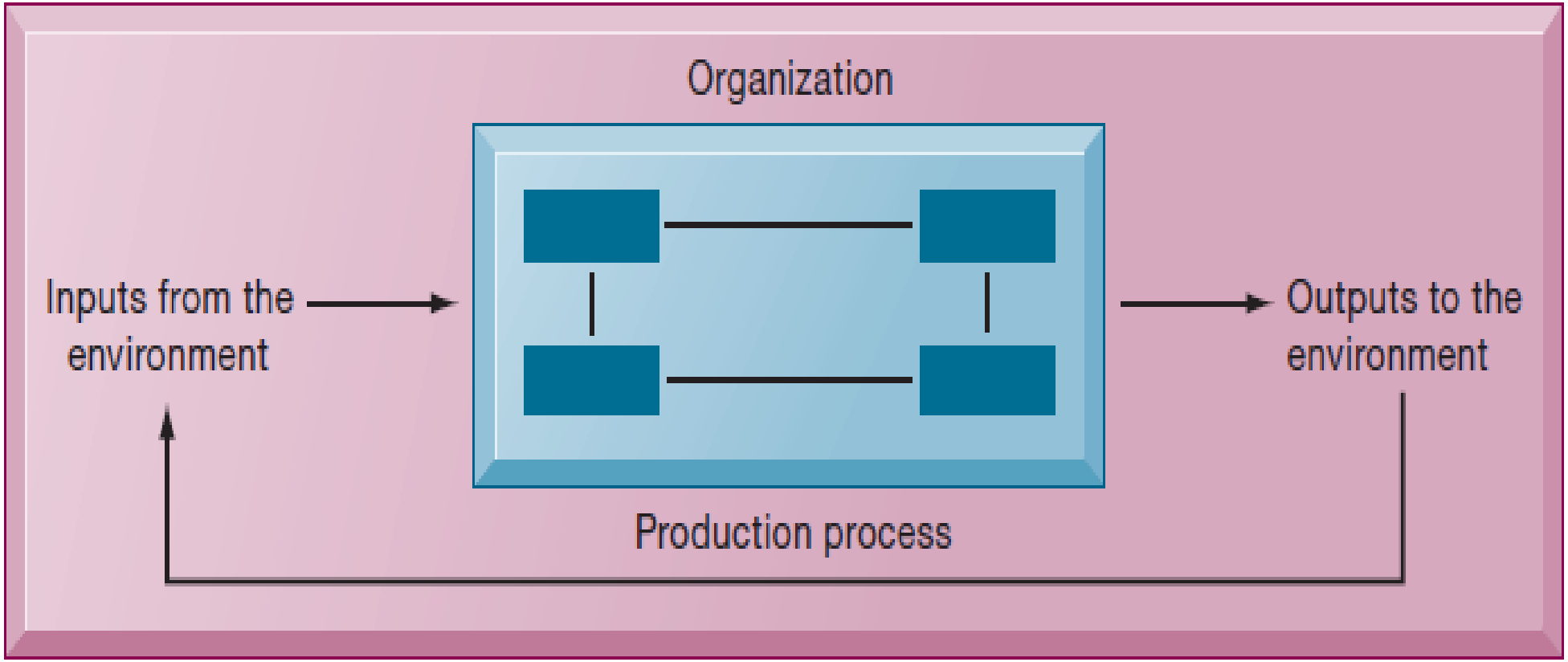
What is an Organization

Technical definition

- An organization is a
  - Stable
  - formal social structure
  - that takes resources from the environment
  - and processes them to produce outputs

- A formal legal entity with internal rules and procedures, as well as a social structure

What is an Organization



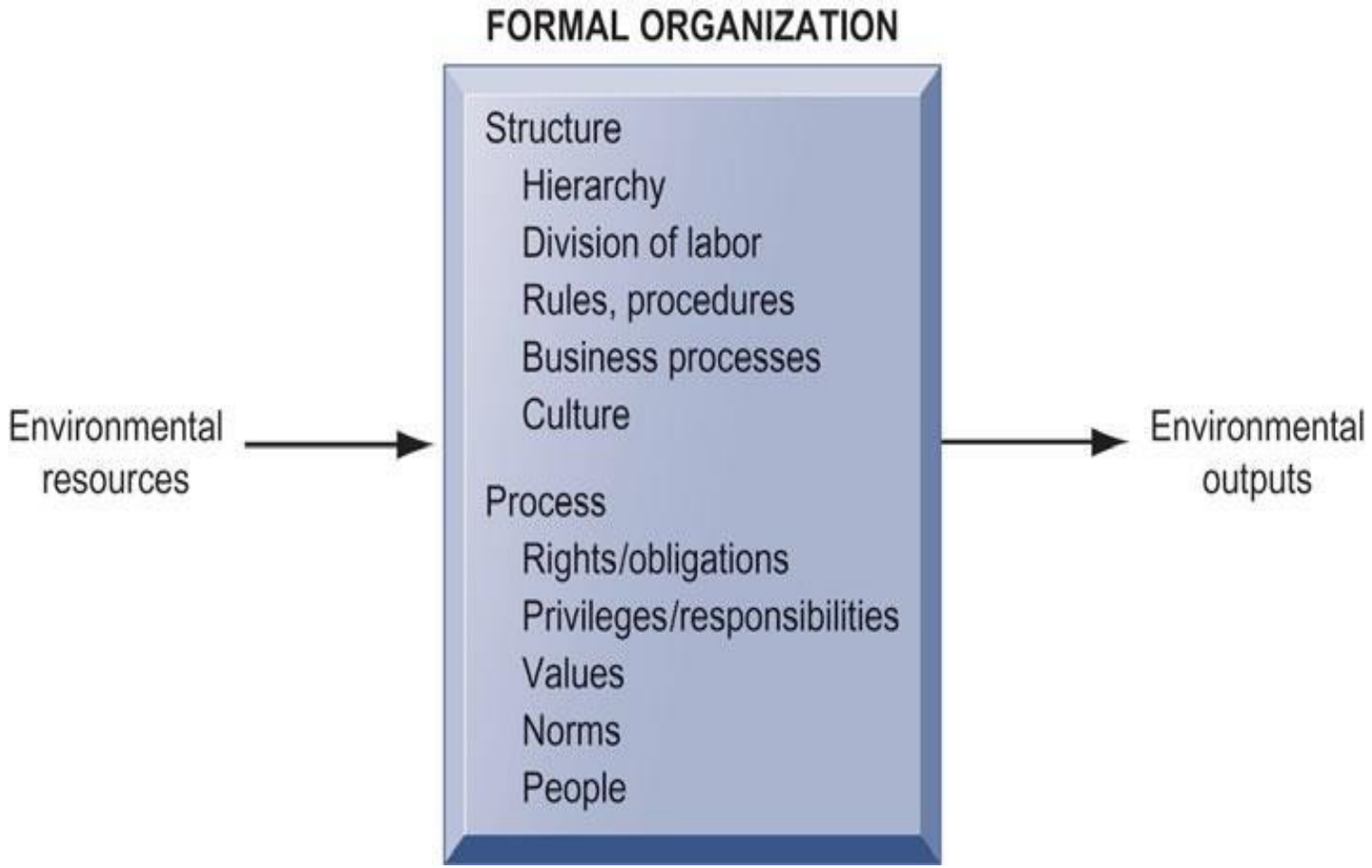
Technical Definition

What is an Organization

Behavioral definition

- A more realistic behavioral definition of an organization is that
  - it is a collection of rights
  - Privileges
  - Obligations
  - and responsibilities
  - that is delicately balanced over a period of time
  - through conflict and conflict resolution

What is an Organization



Behavioral Definition

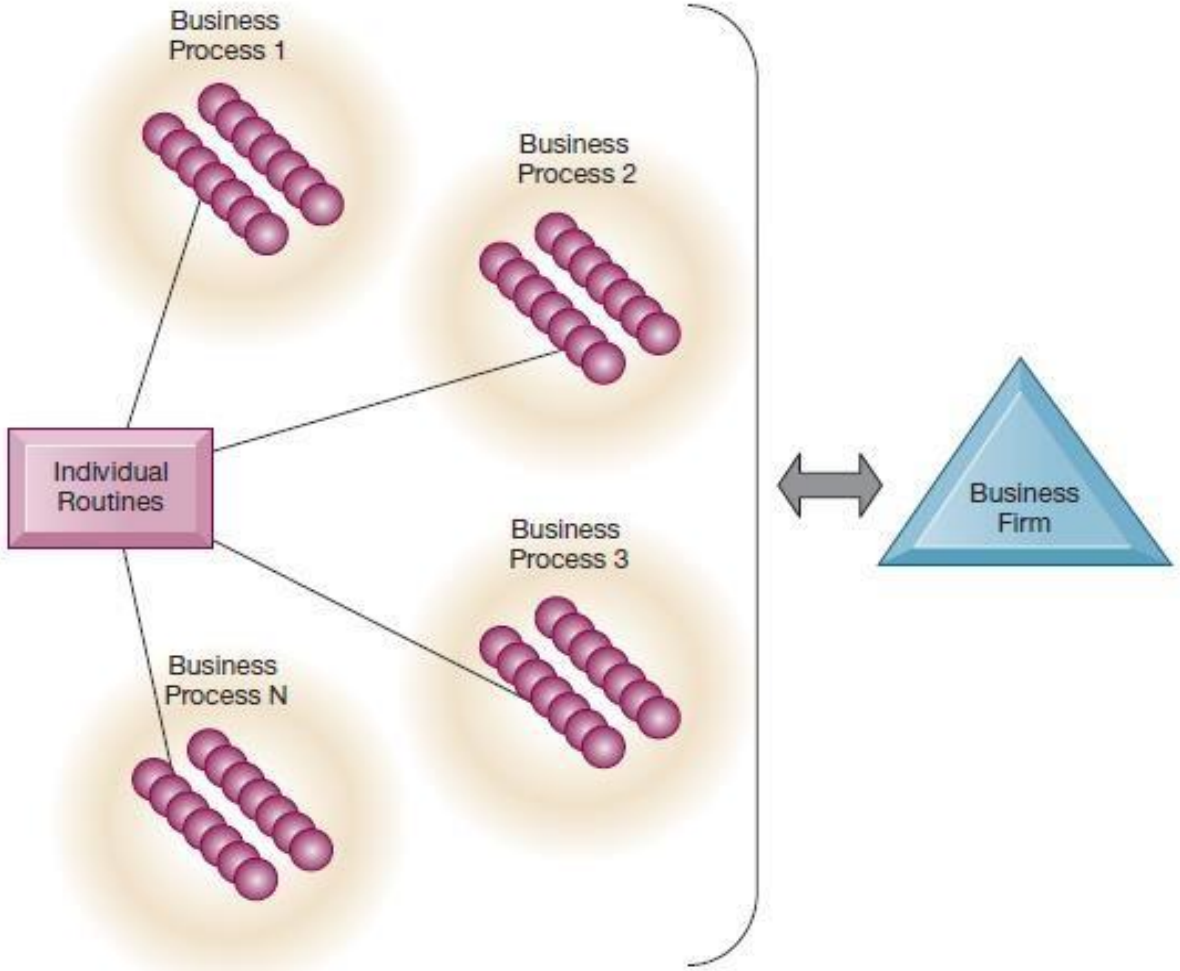
Features of Organizations

- Use of hierarchical structure
- Accountability, authority in system of impartial decision making
- Adherence to principle of efficiency
- Routines and business processes
- Organizational politics, culture, environments and structures

Routines and Business Processes

- Routines (Standard Operating Procedures)
  - Precise rules, procedures, and practices developed to cope with virtually all expected situations
- Business processes:
  - Collections of routines
- Business firm:
  - Collection of business processes

Routines and Business Processes and Firms



Routines

↓

Collections of routines

(Business Processes)

↓

Collection of business processes

(Business Firm)

New information system applications require that individual routines and business processes change to achieve high levels of organizational performance.

Organizational Politics

- Divergent viewpoints * lead to political struggle, competition, and conflict
- Political resistance greatly hampers organizational change

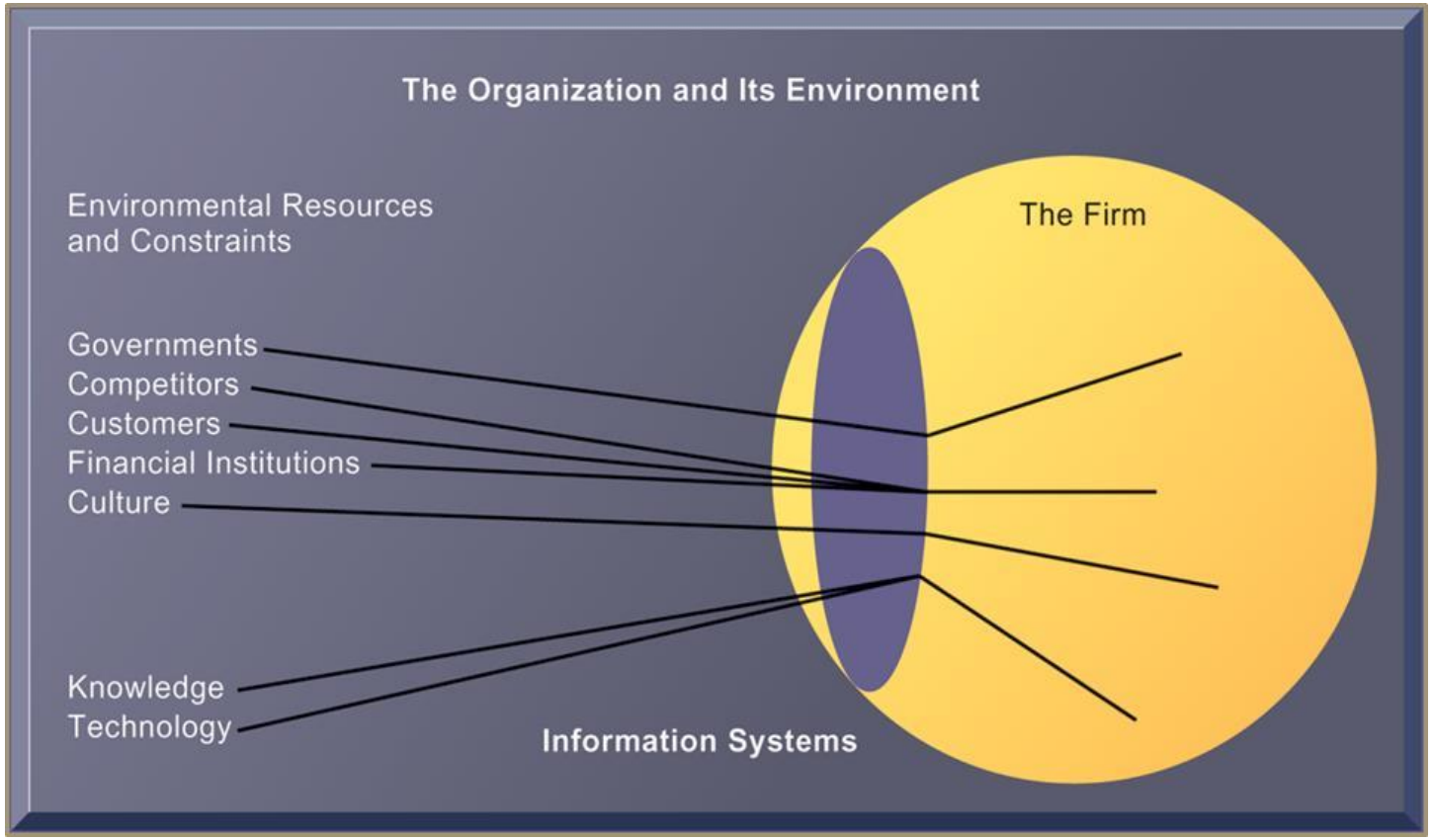
Organizational Culture

- All organizations have bedrock, unassailable, unquestioned (by the members)assumptions that define their goals and products.
- Organizational culture encompasses this set of assumptions about
  - what products the organization should produce
  - how it should produce them
  - where,and for whom.

Organizational Environments

- Organizations and environments have a reciprocal relationship
- Organizations are open to, and dependent on, the social and physical environment
- Organizations can influence their environments
- Environments generally change faster than organizations
- Information systems can be an instrument of environmental scanning, act as a lens

Organizational Environments



The environment affects what an organization can do, but the organization can also change or influence its environment. Information Technology helps the organization understand what is happening around it and respond to those changes.

Disruptive Technologies

- Technology that brings about sweeping change to businesses, industries, markets
- Examples:
  - Personal computers, word processing software, the Internet, the PageRank algorithm
- First movers and fast followers
  - First movers – inventors of disruptive technologies
  - Fast followers – firms with the size and resources to capitalize on that technology

Disruptive Technologies Examples

- Microprocessor chips (1971)
  - Thousands and eventually millions of transistors on a silicon chip
- Personal computers (1975)
  - Small, inexpensive, but fully functional desktop computers
- Digital photography (1975)
  - Using CCD (charge-coupled device) image sensor chips to record images

Disruptive Technologies Examples

- World Wide Web (1989)
  - A global database of digital files and "pages" instantly available
- Internet music, video, TV services (1998)
  - Repositories of downloadable music, video, and TV broadcasts on the web
- PageRank algorithm
  - A method for ranking web pages in terms of their popularity to supplement web search by key terms
- Software as web service
  - Using the Internet to provide remote access to online software

Organizational Structure

- Entrepreneurial structure
  - Small start-up business
- Machine bureaucracy
  - Midsize manufacturing firm
- Divisionalized bureaucracy
  - Fortune 500 firms, such as General motors
- Professional bureaucracy
  - Law firms, school systems, hospitals
- Adhocracy
  - Consulting firms

The Impact Of Information Systems On Organizations

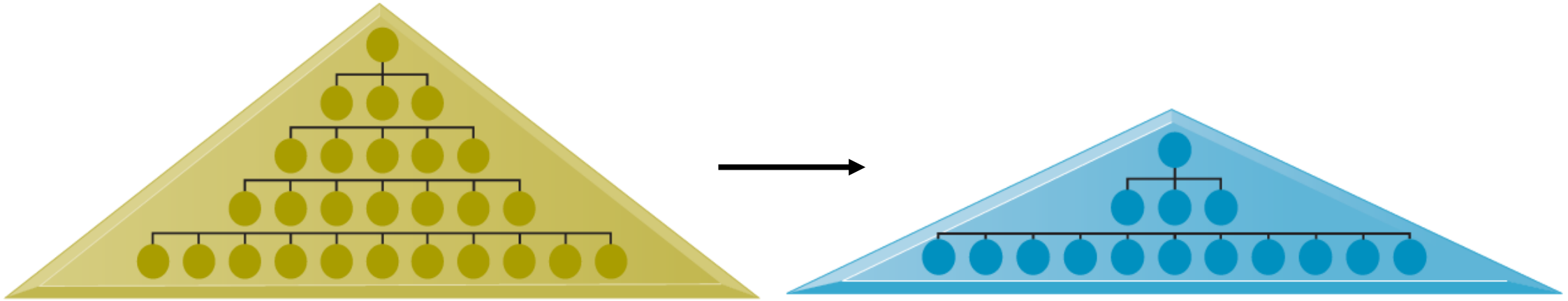
- Economic Impacts
- Organizational and Behavioral Impacts
- Impact of the Internet
- Implications for the Design and Understanding of Information Systems

Economic Impacts

- IT changes relative costs of capital and the costs of information
- Information systems technology is a factor of production, like capital and labor
- IT affects the cost and quality of information and changes economics of information
  - Information technology helps firms contract in size because it can reduce transaction costs (the cost of participating in markets)
    - Outsourcing

Organizational and Behavioral Impacts

- IT Flattens Organizations
  - Decision making pushed to lower levels
  - Fewer managers needed (IT enables faster decision making and increases span of control)



A traditional hierarchical organization with many levels of management

An organization that has been "flattened" by removing layers of management

Organizational and Behavioral Impacts

- Postindustrial Organizations
  - Organizations flatten because in postindustrial societies, authority increasingly relies on knowledge and competence rather than formal positions

Organizational and Behavioral Impacts

- Organizational resistance to change
  - Information systems become bound up in organizational politics because they influence access to a key resource - information
  - Information systems potentially change an organization's structure, culture, politics, and work
  - Most common reason for failure of large projects is due to organizational and political resistance to change

Impact of the Internet

- The Internet increases the accessibility, storage, and distribution of information and knowledge for organizations.

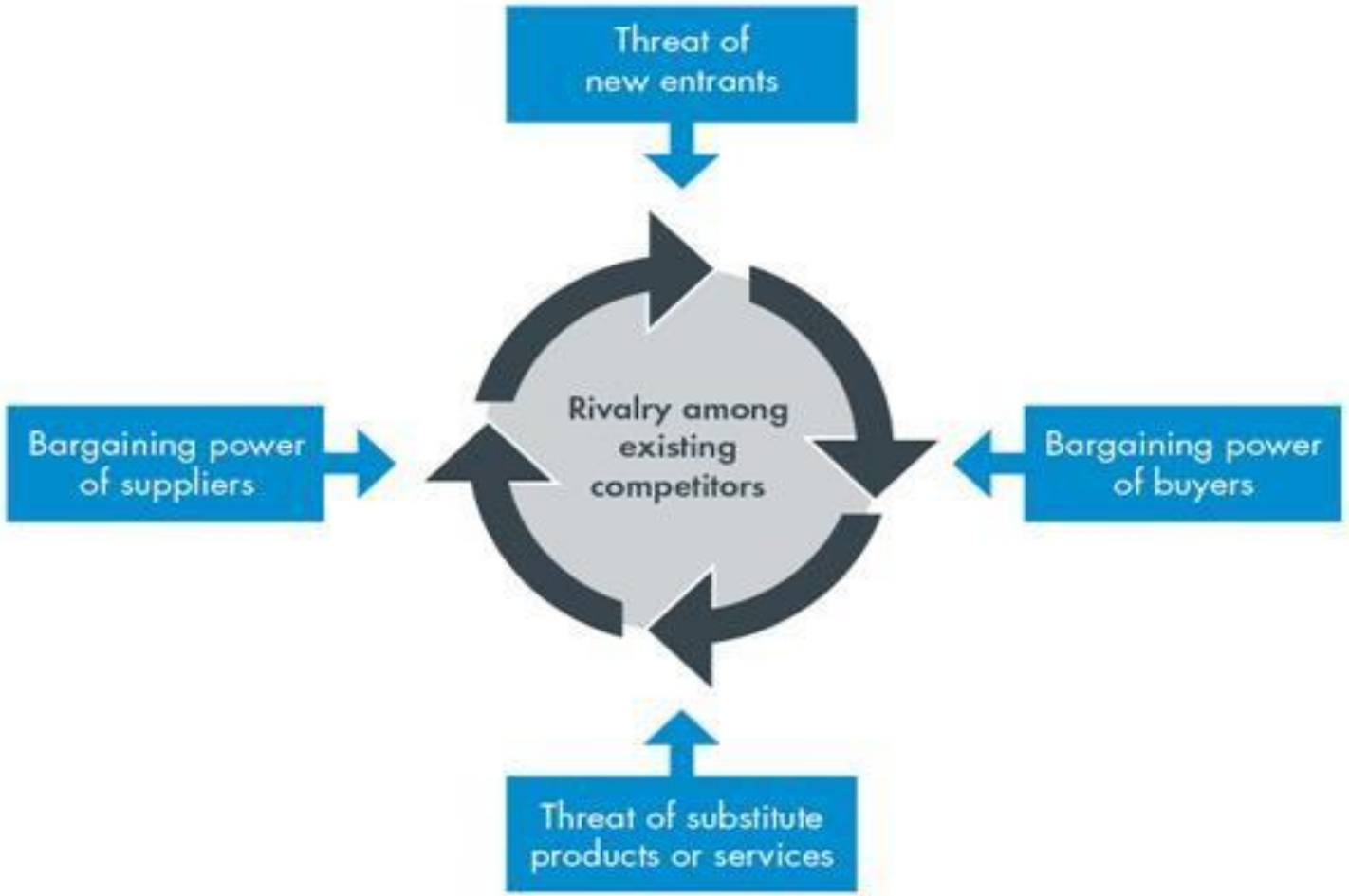
- Internet is capable of dramatically lowering the transaction and agency costs facing most organizations.

- Key component of the firms IT infrastructures

Implications for the Design and Understanding of Information Systems

- The environment in which the organization must function
- The structure of the organization: hierarchy, specialization, routines, and business processes
- The organization's culture and politics
- The type of organization and its style of leadership
- The principal interest groups affected by the system and the attitudes of workers who will be using the system
- The kinds of tasks, decisions, and business processes that the information system is designed to assist

Porter's Five Forces Model



Porter's Five Forces Model

1. Threats of New Entrants
   - Depends on barriers to entry

Barriers to Entry
- Economies of Scale
- Product Differentiation
- Capital Investments
- Switching Cost
- Government Policy
- Patent Right
- Brand Preference/Customer Loyalty

Porter's Five Forces Model

2. Rivalry among Existing Competitors
   - Concentration Ratio
   - Industry Growth
   - Relationship between Competitors
   - Competencies
     - Resources, Leadership, Brand, Innovation
   - Exist Barriers
     - Redundancy Payments, Effect of withdrawal on other operations, Reluctant of management to accept the defeat, Government pressure

Porter's Five Forces Model

3. Threats of Substitutes
   - Satisfy same need

Depend on
- Price
- Relative Performance
- Switching Cost
- Change of need/ taste

Porter's Five Forces Model

4. Bargaining Power of Customers/ Buyers

- Product Variety
- Product Differentiation
- Pricing
- Income of Buyers
- Buyers' knowledge of Alternative Products
- Volume Purchased
- Price Sensitivity of Buyers

Porter's Five Forces Model

5. Bargaining Power of Suppliers

- Number of Suppliers
- % of Suppliers' Business to the Company
- Market Share,Size of Suppliers
- Importance of the Suppliers Product to the Buyer

Information System Strategies for Dealing with Competitive Forces

- Low-cost leadership
  - Use information systems to produce products and services at a lower price than competitors while enhancing quality and level of service

- Product differentiation
  - Use information systems to differentiate products, and enable new services and products

Information System Strategies for Dealing with Competitive Forces

- Focus on market niche
  - Use information systems to enable a focused strategy on a single market niche;specialize

- Customer and supplier intimacy
  - Use information systems to develop strong ties and loyalty with customers and suppliers

Impact of the Internet on Competitive Forces and Industry Structure

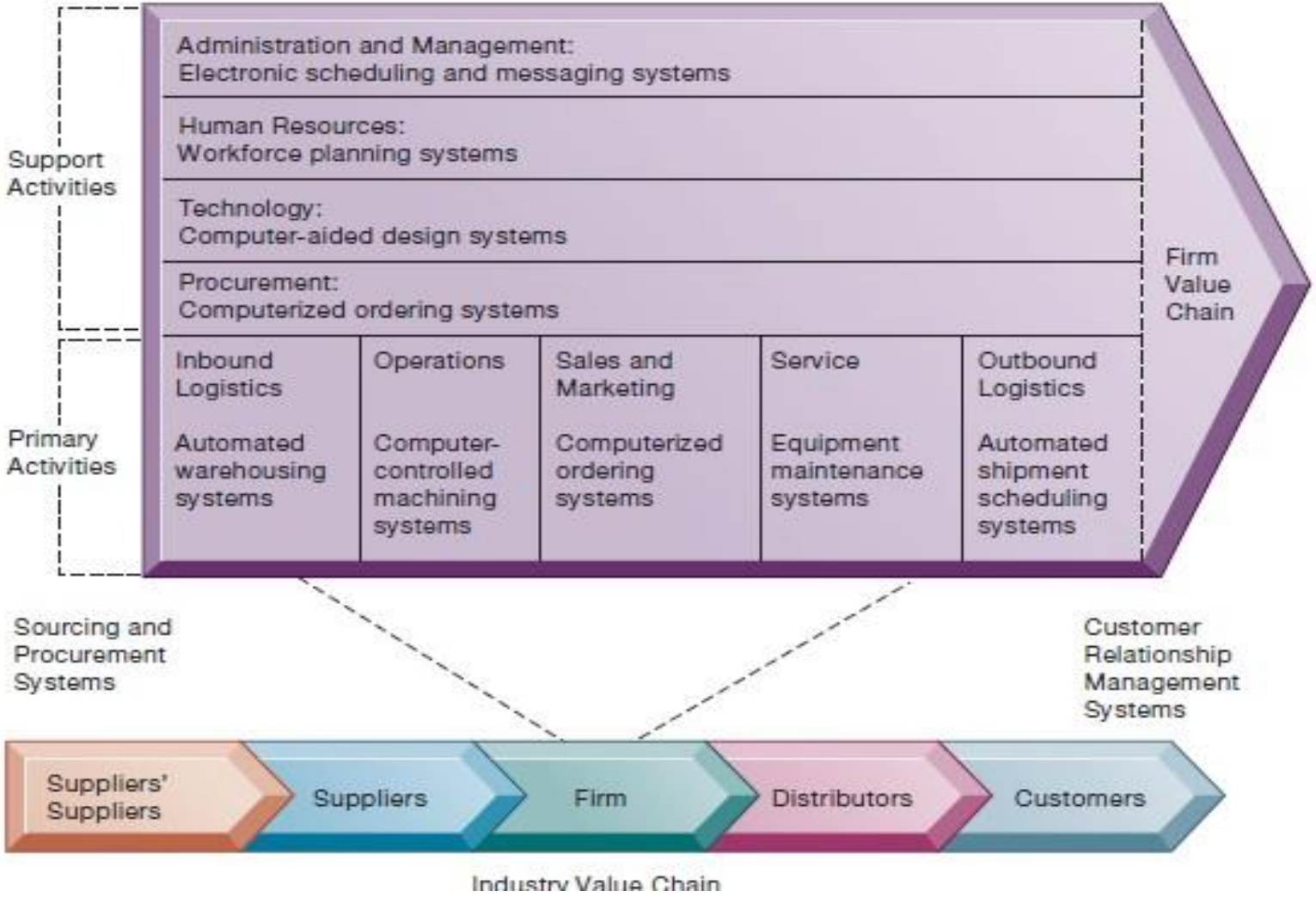
- Substitute products or services
  - Enables substitutes to emerge with new approaches to meeting needs and performing functions
- Customers' bargaining power
  - Availability of global price and product information shifts bargaining power to customers
- Suppliers' bargaining power
  - Procurement over the Internet tends to raise bargaining power over suppliers;
  - suppliers can also benefit from reduced barriers to entry and from the elimination of distributors and other intermediaries standing between them and their users

Impact of the Internet on Competitive Forces and Industry Structure

- Threat of new entrants
  - Reduces barriers to entry, such as the need for a sales force, access to channels, and physical assets;
  - provides a technology for driving business processes that makes other things easier to do

- Positioning and rivalry among existing competitors
  - Widens the geographic market, increasing the number of competitors and reducing differences among competitors;
  - makes it more difficult to sustain operational advantages;
  - puts pressure to compete on price

Porter's Value Chain Analysis



The Value Chain Model

Porter's Value Chain Analysis

- Porter says 'Competitive advantage arise out of the way in which the firms organize themselves and perform its activities"

- VCA analysis' the means by which a firm creates value into its products in the perspective of the end user"

- Porter says that management should analyze their business in terms of how it generates value for customers and shareholders

Porter's Value Chain Analysis

- Inbound Logistics
  - Receiving, Storing, Stock Control of Raw Materials
- Operations
  - Conversion of Input to Finish Goods
  - Production System
  - Capacity Utilization
- Outbound Logistics
  - Storing, Testing, Packaging, and Distribution of the Finished Goods
- Marketing & Sales
  - Promotional Activities
  - Informing customers and persuading them to buy

Porter's Value Chain Analysis

- [After Sales] Services
  - Maintenance, Repairing, Guarantees, Advisory Services
- Procurement
  - Purchase of Inputs for Primary Activities
- Technological Developments[R & D]
  - Product and Process Design
  - Continuous Improvement
  - Innovation
- Human Resources
  - Recruitment, Training and Development, Motivation and Rewarding

Porter's Value Chain Analysis

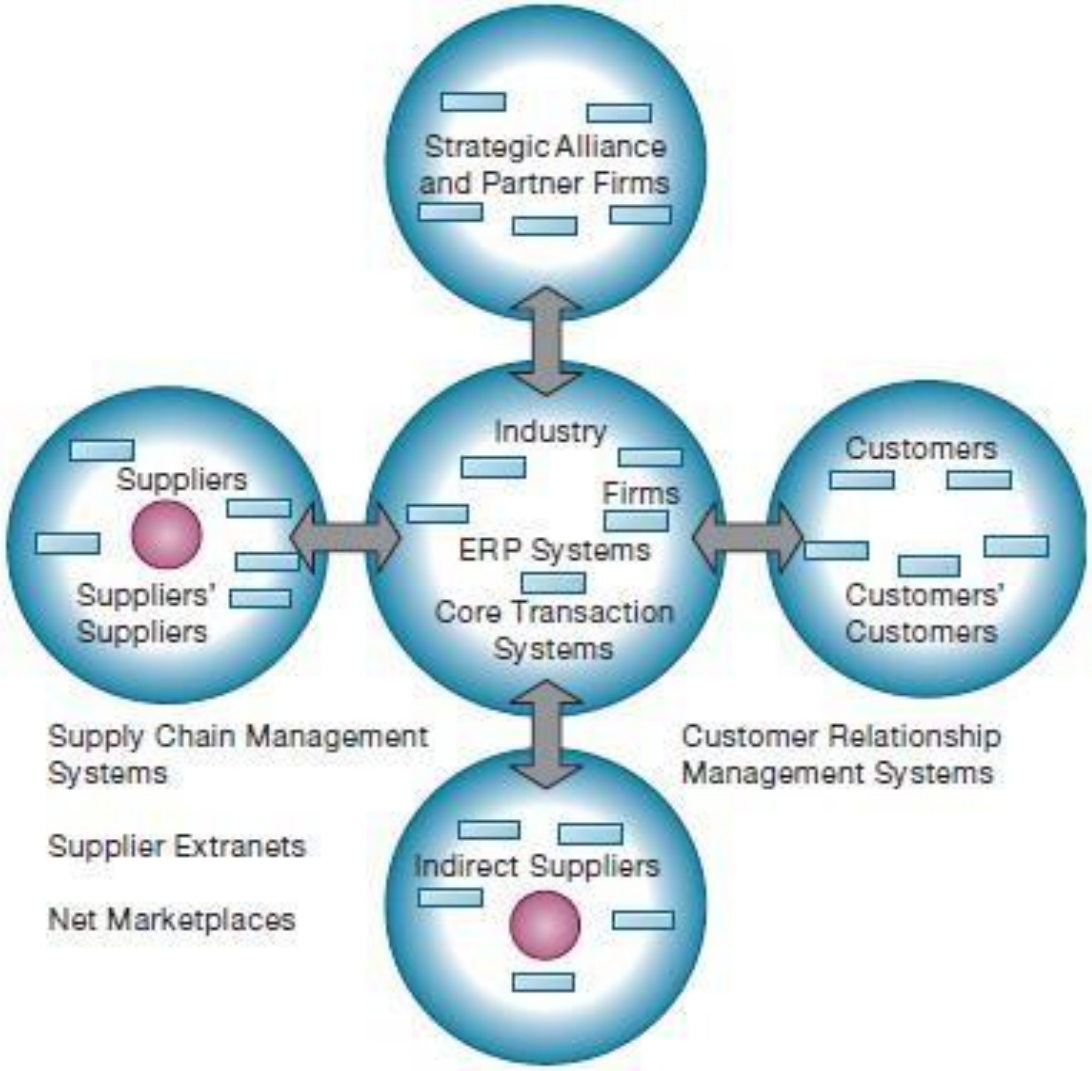
- Firm Infrastructure
  - Structure
  - Culture
  - Management Planning
  - Finance

Extending the Value Chain: The Value Web

- Value web is a collection of independent firms that use information technology to coordinate their value chains to produce a product or service for a market collectively.

- It is more customer driven and operates in a less linear fashion than the traditional value chain.

Extending the Value Chain: The Value Web



The value web is a networked system that can synchronize the value chains of business partners within an industry to respond rapidly to changes in supply and demand.

How Information Systems Can Improve the Overall Performance of Business Units

- Synergies
- Core Competencies
- Network-Based Strategies

What are the challenges posed by strategic information systems, and how should they be addressed?

- Sustaining Competitive Advantage
- Aligning IT with Business Objectives
- Managing Strategic Transitions

Summary

- What is an Organization
  - Technical and Behavioral definitions

- Features of Organizations
- Porter's Five Forces Model
- Porter's Value Chain Analysis

Questions ???



Thank you…